

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE																		
Supplementary Examination – Summer 2022																		
Course: B. Tech.	Branch : Computer Science Engineering	Semester : IV																
Subject Code & Name: BTCOE406B Numerical methods																		
Max Marks: 60	Date:	Duration: 3 Hr.																
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.																		
	(Level/CO)	Marks																
Q. 1	Solve Any Two of the following.																	
A)	Find a real root of the equation $x^3 - x - 11 = 0$ by using bisection method correct to four place of decimals.	CO-2 6																
B)	Find by Regula falsi method the positive root of $x^2 - \log_{10} x - 12 = 0$ correct to three place of decimals.	CO-2 6																
C)	Evaluate $\sqrt{12}$ to four places of decimals by Newton-Raphson method.	CO-2 6																
Q.2	Solve the following.																	
A)	Solve the following system of equations using Gauss Seidel iteration method. $6x + 15y + 2z = 72, x + y + 54z = 110, 27x + 6y - z = 85.$	CO-1 4																
B)	Solve the following system of equations by Jacobi method. $28x + 4y - z = 32, x + 3y + 10z = 24, 2x + 17y + 4z = 35.$	CO-1 4																
C)	Solve the following system of equations using Gauss elimination method. $x + y + z = 9, 2x - 3y + 4z = 13, 3x + 4y + 5z = 40.$	CO-1 4																
Q. 3	Solve Any Two of the following.																	
A)	From the following data obtain the first and second derivatives of $y = \log_e x$ at $x = 500$ and $x = 550$ <table><tr><td>x</td><td>500</td><td>510</td><td>520</td><td>530</td><td>540</td><td>550</td></tr><tr><td>$y = \log_e x$</td><td>6.2146</td><td>6.2344</td><td>6.2538</td><td>6.2729</td><td>6.2916</td><td>6.3099</td></tr></table>	x	500	510	520	530	540	550	$y = \log_e x$	6.2146	6.2344	6.2538	6.2729	6.2916	6.3099	CO-3 6		
x	500	510	520	530	540	550												
$y = \log_e x$	6.2146	6.2344	6.2538	6.2729	6.2916	6.3099												
B)	Find the gradient of the road at the middle point of the elevation above a datum line of seven points of a road which are given below: <table><tr><td>x</td><td>0</td><td>300</td><td>600</td><td>900</td><td>1200</td><td>1500</td><td>1800</td></tr><tr><td>y</td><td>135</td><td>149</td><td>157</td><td>183</td><td>201</td><td>205</td><td>193</td></tr></table>	x	0	300	600	900	1200	1500	1800	y	135	149	157	183	201	205	193	CO-3 6
x	0	300	600	900	1200	1500	1800											
y	135	149	157	183	201	205	193											
C)	(i) Show that $\Delta \log f(x) = \log \left[1 + \frac{\Delta f(x)}{f(x)} \right]$ and (ii) Find $\Delta^n \sin x$ taking $h = 1$.	CO-3 6																

Q.4	Solve Any One of the following.		
A)	Evaluate $\int_0^1 \frac{dx}{1+x}$ using (i) Trapezoidal rule (ii) Simpson's one third rule (iii) Simpson's three eight rule. Find the error in each method by comparing with the actual integration up to 4 places of decimals. Take $h = \frac{1}{6}$.	CO-1	12
B)	Evaluate $I = \int_4^{5.2} \log_e x \, dx$ using Trapezoidal , Simpson's both rule .	CO-1	12
Q. 5	Solve Any One of the following.		
A)	Solve the differential equation $\frac{dy}{dx} = \frac{1}{(x+y)}$, $x_0 = 0$, $y_0 = 1$ for the interval (0, 1) choosing $h = 0.5$ by Runge –Kutta method of fourth order.	CO-1	12
B)	Determine the value of y when $x = 0.1$ given that $y(0) = 1$ and $\frac{dy}{dx} = x^2 + y$ by Euler's modified method taking $h = 0.05$.	CO-1	12
	*** End ***		